

Senior Network Engineer | Delivered Automation Platform Validating 1,000+ Circuits in Under 2 Minutes**IMPACT SNAPSHOT: 90,000-Mile IP/MPLS Backbone Design — 25 Multi-Metro PoP Deployments — \$3M High-Density Wireless Overhaul**

Senior network engineering leader with 15+ years designing, scaling, and securing high-capacity IP/MPLS backbones and multi-site enterprise networks across carrier-grade and cloud-integrated environments. Recognized for uniting protocol expertise (BGP, OSPF, IS-IS, MPLS) with advanced automation skills in Python, Ansible, and CI/CD tooling to accelerate network provisioning, strengthen uptime, and optimize performance at a national scale. Verifiable success leading multi-million-dollar infrastructure builds, streamlining migration and incident response processes, and delivering high-performance connectivity supporting critical operations for service providers, enterprises, and public-sector clients.

AREAS OF EXPERTISE

- Cloud Network Architecture
- High-Performance Network Optimization
- Network Automation (Python, Ansible)
- Troubleshooting and Root Cause Analysis
- Data Center Interconnect (DCI)
- Underlay and Overlay Networking
- Network Virtualization (VRF)
- Network Infrastructure Monitoring
- Large-Scale Network Design
- BGP, OSPF, and IS-IS Routing
- Peering and Transit Engineering
- MPLS

TECHNICAL SKILLS

Networking and Protocols: MPLS, RSVP, Traffic Engineering (TE), VPLS, LDP, T-LDP, IS-IS, OSPF, BGP, MP-BGP, Layer 2/3 VPNs, VRF, QoS, MACSec, PEAP, EAP-TLS, Q-in-Q, G.8032, LACP, VPC, MSTP, DWDM, OAM, SNMP | **Vendors:** Juniper, Cisco, Nokia, Ciena, Extreme, Infoblox
Coding and Automation: Python, Ansible, Git, VSCode, Bash, SQL, SQLite, gRPC, REST, C, C++, Assembler, HTML/CSS | **Systems and Tools:** Grafana, Jenkins, NSO, Notion, Obsidian, Confluence, Power BI, Kentik, Elastic, Splunk, Cisco WAE, Blue Planet ROA, SevOne, SolarWinds

CAREER HIGHLIGHTS**Wireless Network Design**

Spearheaded a \$3M high-density wireless overhaul across Boston's convention venues, orchestrating vendor coordination, RF design, and deployment of 700+ 802.11ac/ax access points to enable 504% higher throughput, seamless roaming for 57K+ concurrent users, and zero post-event performance complaints.

Backbone Engineering

Engineered and launched a \$500K Juniper MX backbone integrating VPLS and L3VPN to consolidate three isolated networks, slashing OpEx by 37% and CapEx by 35%, delivering sub-50ms failover, and streamlining service provisioning for 7 sites.

Network Infrastructure Modernization

Designed and implemented a \$1.5M Juniper EX distribution and access layer upgrade, replacing legacy switching with high-throughput architecture to support 700 Gbps wireless backhaul, reduce latency by 66%, and scale capacity for 35K-attendee digital events.

PROFESSIONAL EXPERIENCE**Senior Network Engineer, CROWN CASTLE / ZAYO**

April 2015–July 2026

Architect and scale high-capacity service provider networks across a 90,000-mile fiber backbone, delivering IP/MPLS, SD-WAN, and cloud services to enterprise customers. Lead automation-driven transformation using Python, Ansible, and Jenkins to accelerate circuit migrations, fine-tune infrastructure performance, and increase network resilience in complex multi-vendor environments.

- Built a Python-driven automation platform to **execute post-migration checks on 1,000+ circuits in under two minutes**, auto-generate validation reports, and protect high-risk network cutovers, eliminating outages and strengthening quality assurance nationwide.
- **Directed end-to-end deployment of 25 Points of Presence in major U.S. metros**, overseeing fiber design, capacity planning, and multi-vendor integration to expand network reach, unlock high-revenue markets, and ignite enterprise and cloud service growth.
- Led multi-department incident response to network outages, driving root cause analysis and architectural redesigns **improving backbone uptime by 33% and reducing mean-time-to-repair (MTTR) to under 5 minutes**, stabilizing service delivery for enterprise customers.
- Developed a Power BI analytics suite mapping topology, equipment, and circuit health for an 8,000-customer acquired network, enabling leadership to **decrease migration planning cycles by 25%**, optimize sparing strategies, and **minimize service risk across 13 markets**.

Network Manager, MASSACHUSETTS CONVENTION CENTER AUTHORITY

April 2014–April 2015

Assembled and managed a team of network engineers and system administrators to expand, modernize, and secure enterprise backbone and infrastructure systems. Oversaw multimillion-dollar capital projects, technology refreshes, and process redesigns shifting operations from reactive fixes to proactive strategy, earning executive trust and restoring service delivery benchmarks.

- **Led a \$350K data center overhaul**, consolidating five legacy networks into a Juniper QFX CLOS leaf-spine fabric to **boost throughput by 566%, decrease latency by 4ms, reduce device footprint by 33%, and lower total cost of ownership by 31%**.
- **Deployed a \$700K Juniper EX distribution and edge architecture delivering 10 Gbps low-latency connectivity** for next-gen audio systems, enabling the Boston Convention Center to host 12% more high-tech events and match top-tier U.S. venues.
- Introduced a peer-reviewed change control process with approval gates, rollback workflows, and staged testing, **declining unplanned outages by 20%, shortening maintenance windows by 3 hours**, and ensuring 100% compliance.
- Built a 3-year infrastructure roadmap and an interactive resource planning system tracking budgets, milestones, and dependencies, **elevating on-time project delivery from 15% to 86%** and facilitating proactive capacity and staffing decisions.

Senior Network Engineer, MASSACHUSETTS CONVENTION CENTER AUTHORITY

February 2012–April 2014

Drove major modernization of enterprise networking through the design and deployment of IP/MPLS systems, Juniper-based architectures, and custom visibility tools to improve uptime and service quality. Championed smooth integration across data centers, backbone networks, and wireless platforms while unifying operations through automation and centralized management.

- **Rescued a stalled \$500K network overhaul** by deploying Juniper ERX1400 and SRC2000 to enable BGP peering and branded wireless portals, driving **25% Wi-Fi uptime improvement** and **generating 20% more revenue from guest internet services**.
- **Architected a \$60K unified network management platform** by integrating SolarWinds, NetFlow, LEM, RADIUS, and NTP into a single pane of glass, **cutting provisioning time by 66%, reducing operational costs by \$12K annually**, and **improving fault isolation speed by 59%**.
- Created a custom SQL-integrated web tool for automated switch port provisioning across 7 sites, **slashing fulfillment times from 2 days to 1 hour, eliminating 65% of backlog tickets, and freeing 15 staff hours monthly**.

Resident Engineer, JACOBS ENGINEERING

February 2011–February 2012

Designed and implemented enterprise-class networking solutions in critical infrastructure environments, combining high-availability switching, advanced access controls, and automation to strengthen performance, security, and uptime. Developed custom scripts and led multi-site migrations to replace legacy systems and maintain operational continuity under tight delivery pressures.

- **Streamlined password and configuration updates for 250+ network devices** with a custom Visual Basic and Perl automation suite, shrinking a multi-day process to minutes, eliminating manual error, and **ensuring 100% compliance with security audit protocols**.
- Devised a VMware disaster recovery framework leveraging Cisco 3021 switches and segmented VLANs to facilitate seamless cross-site vMotion, **minimizing failover time from 3 hours to 5 minutes** and **protecting 234 systems from service disruption**.
- Revamped a 32-site surveillance network by redesigning the IP scheme and deploying PIM routing, eradicating outage incidents, **reducing false police dispatches by 98%**, and restoring uninterrupted monitoring across ports, airports, and transit terminals.

Network Engineer II, PARTNERS HEALTHCARE

April 2010–December 2010

Engineered enterprise network solutions across hospital and office campuses by delivering MPLS and Layer 3 VPN services, Cisco and Nexus switching, and wireless infrastructure supporting over 80K users.

- **Directed wireless design for a 14-floor, 275,000 sq/ft hospital**, utilizing AirMagnet survey data to overcome structural and RF challenges, achieve 90% voice-grade 802.11 a/b/g coverage, and **influence uninterrupted mobility for 1200 clinical staff and 11K devices**.
- Designed high-availability LAN/WAN architecture for 5 hospital expansion sites using Cisco Catalyst and Nexus VPC, **delivering 99% uptime for major clinical applications** and decreasing deployment schedules through targeted mentoring of junior engineers.

Resident Engineer, JACOBS ENGINEERING

May 2009–April 2010

Transformed network operations across multi-site enterprises by standardizing engineering workflows, refining IP schema, and executing large-scale protocol migrations to boost scalability, performance, and security.

- Delivered a multi-site network overhaul spanning four data centers and 30+ remote locations, **increasing backbone capacity by 300%** and **enabling 5000 concurrent surveillance video streams** through high-throughput Enterasys S-Series deployment.
- **Lowered deployment failures by 25%** and **accelerated production rollouts by 30 hours per change** by rebuilding the enterprise change control process with pre- and post-network diagrams, configuration snapshots, and structured acceptance testing.

EARLY CAREER:

Network Engineer, MASSACHUSETTS CONVENTION CENTER AUTHORITY

LAN Administrator, STATE STREET BANK